

Research Statement

Tianqi Xu

Research Vision

My research goal is to develop Vision-Language-Action (VLA) models that reason efficiently and generalize reliably under real-world constraints, including limited computation, delayed inference, and distribution shifts.

Research Experience

My research began in September 2024 under **Prof. Tao He** at **Sichuan University**, focusing on ODE-based medical image segmentation. Continuous-time ODE models offered a flexible formulation of feature evolution, but could incur substantial computational cost. Together with the cross-domain generalization challenges in medical image segmentation, this motivated me to investigate whether ODE-based modeling could be made both compact and robust.

My first authored **AAAI 2026** paper, **CNM-UNet**, introduced a single Continuous Neural Memory ODE block that replaced multiple hierarchical decoder layers, achieving better segmentation accuracy while maintaining efficiency. I also designed a dual self updating strategy to improve cross domain adaptation. I led the project from problem formulation through experimentation and manuscript writing, with guidance from **Prof. He**.

The generalization challenges I encountered in **CNM-UNet** naturally led me to delve deeper into test-time adaptation. With **Prof. Tao He**, I explored continual adaptation, selective parameter updating, and optimizer-state reset, resulting in a paper under review at **Pattern Recognition** and another under review at **ICLR 2027**.

My presentation at **AAAI 2026** in Singapore exposed me to recent advances in embodied AI and multimodal reasoning, sparking my interest in how foundation models can reason and act in the physical world. In March 2026, I joined **Prof. Yuzhang Shang**'s group at the **University of Central Florida** as a remote research intern. I am currently investigating chain-of-thought reasoning and self-supervised reinforcement learning for VLA models. This experience has given me a grounding in the field and established efficient reasoning and deployment-time generalization as my primary research interests.

Proposed Research

Efficient Multimodal Reasoning for VLA

Explicit autoregressive CoT requires sequential token generation before action generation, substantially increasing inference latency and limiting its applicability in real-time closed-loop control. I therefore aim to replace explicit autoregressive reasoning with latent CoT that can be generated in a single forward pass, and develop lightweight supervision to enable efficient reasoning within compact latent representations.

Recent studies and my ongoing experiments suggest that the benefits of combining visual and linguistic CoT are task-dependent, motivating a closer study of their complementary and potentially redundant information. I therefore plan to investigate how these reasoning modalities can be integrated to preserve action-relevant information while reducing redundancy and resolving potential conflicts, with the goal of improving action generation.

Test-Time Self-Improvement for VLA

VLA models are typically fine-tuned on limited offline demonstrations, which can overfit the policy to the training distribution. However, real-world deployment inevitably encounters unseen objects, viewpoints, tasks, and physical conditions, leading to substantial performance degradation in novel environments.

I aim to develop unsupervised test-time adaptation methods that enable VLA models to improve through deployment-time interaction without additional human annotations. I plan to explore this direction under moderate distribution shifts within established benchmarks and more challenging sim-to-real transfer settings. Ultimately, I hope to enable VLA policies to continually adapt and self-improve when encountering previously unseen conditions.

References

[1] Xu et al. "CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation." **AAAI**, 2026.